Architecture

How the BrewTec Employee Onboarding Journey Agent is built: the parts, how data moves between them, the data model and the main design decisions.

Related documents: Workflows · Security and governance · Testing · Monitoring · Deployment

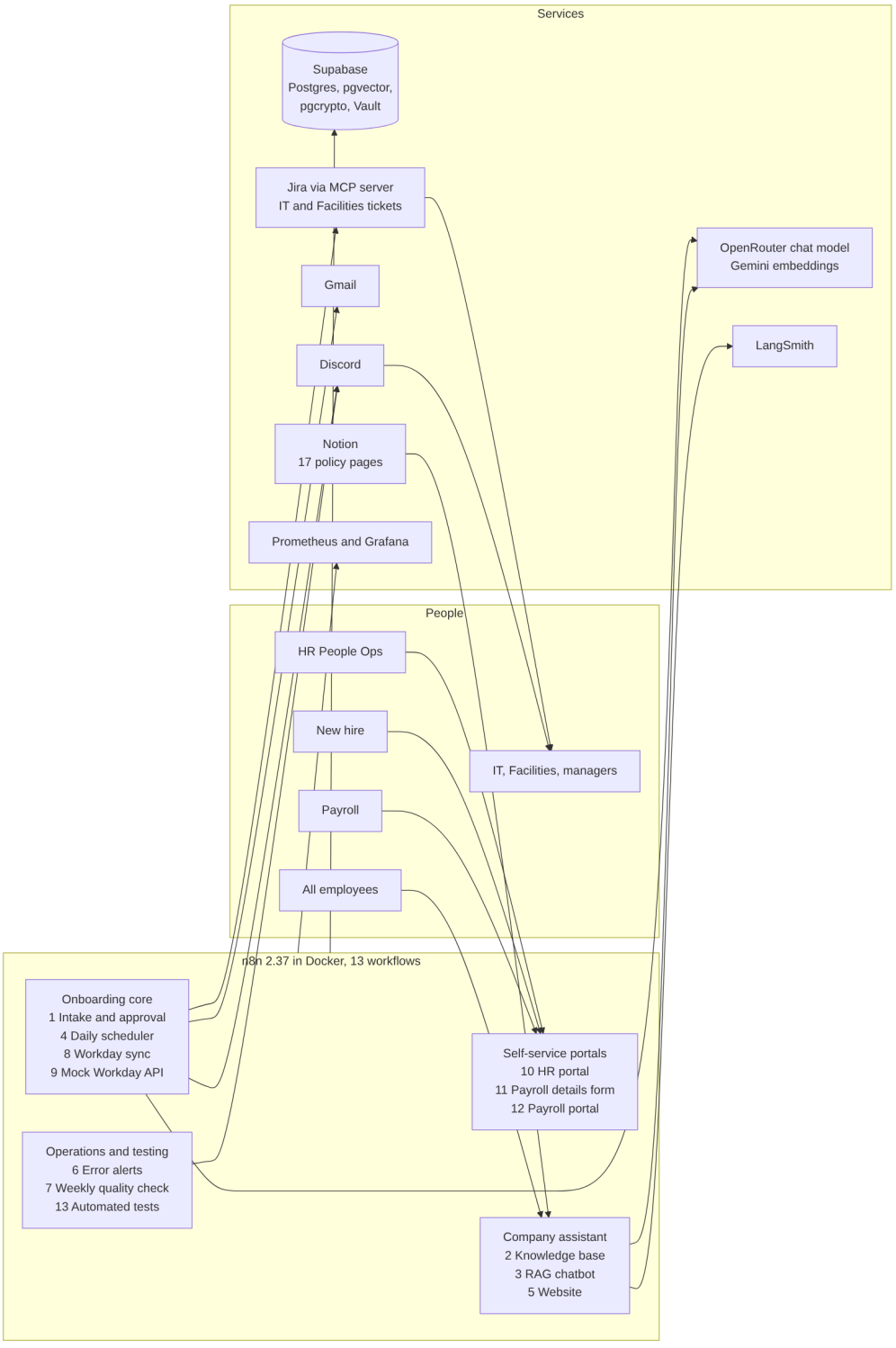
1. Design principle

The AI writes text. Code decides. People approve.

Decision	Made by
Is this hire valid (department, 14 days notice, no duplicate)?	Rules in a Code node
Which 25 tasks, for which team, due when?	Rules in a Code node
Welcome e-mail text	AI draft, checked by code, approved by HR
Does the e-mail go out?	HR, one click in Gmail
Answer to an employee question	AI, only from retrieved documents, checked by code
Who sees bank and tax details?	Payroll only, every view logged

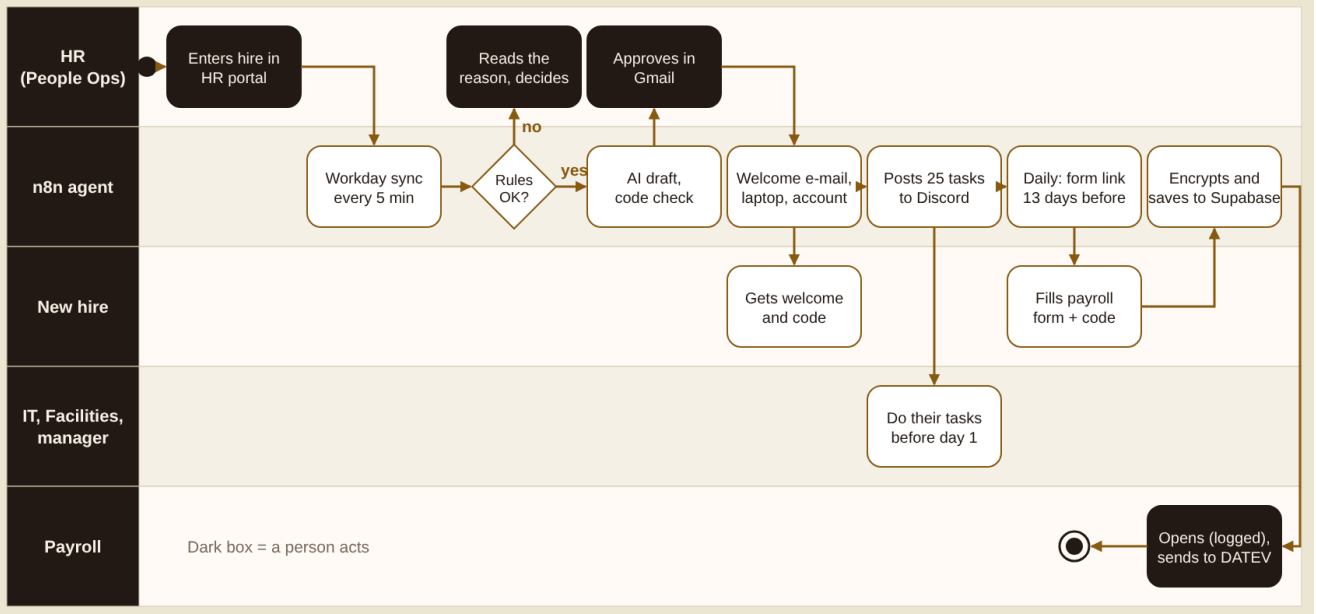
Employment is a high-risk area under the EU AI Act (Annex III). Keeping every decision about a person out of the AI is the core of the design.

2. System overview

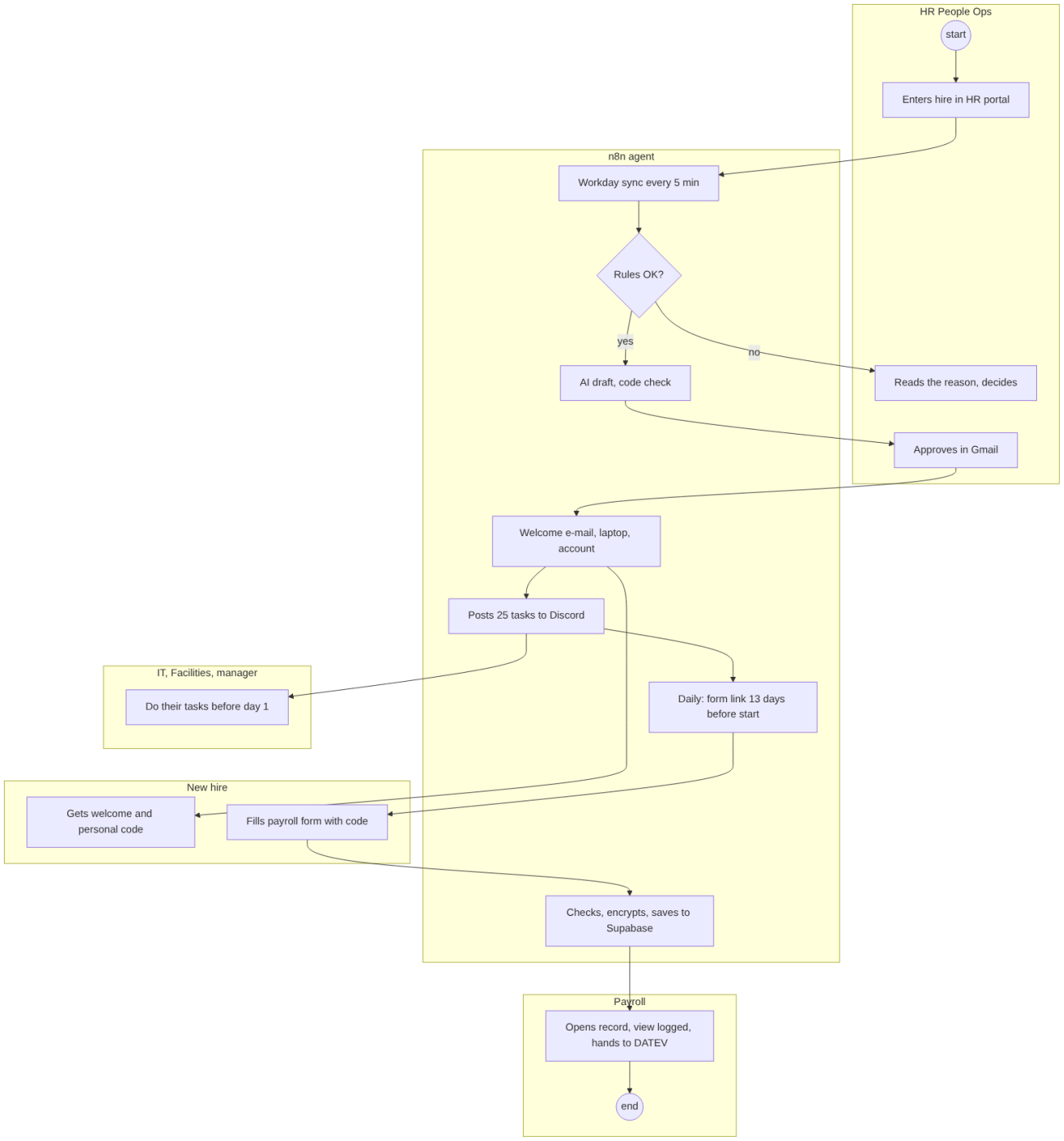


3. The whole flow (UML activity diagram)

One hire from HR entry to the payroll hand-over. Each lane is a role, dark boxes are where a person acts.



The same flow as a Mermaid diagram, which GitHub draws from text:

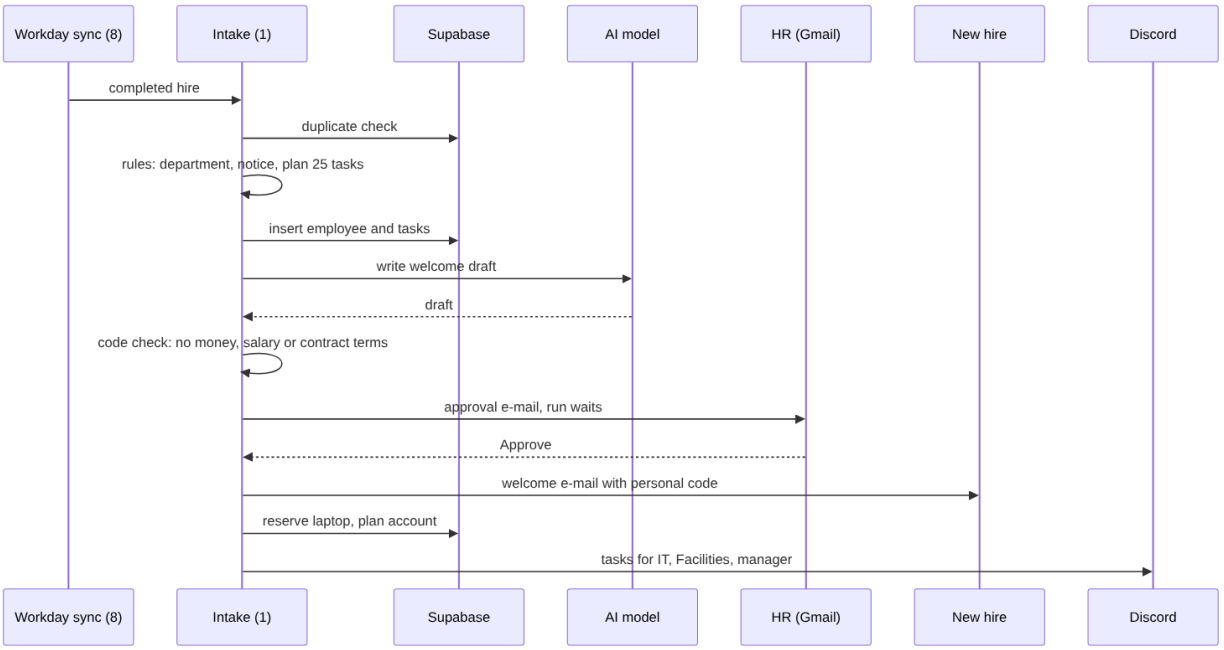


4. Data flows

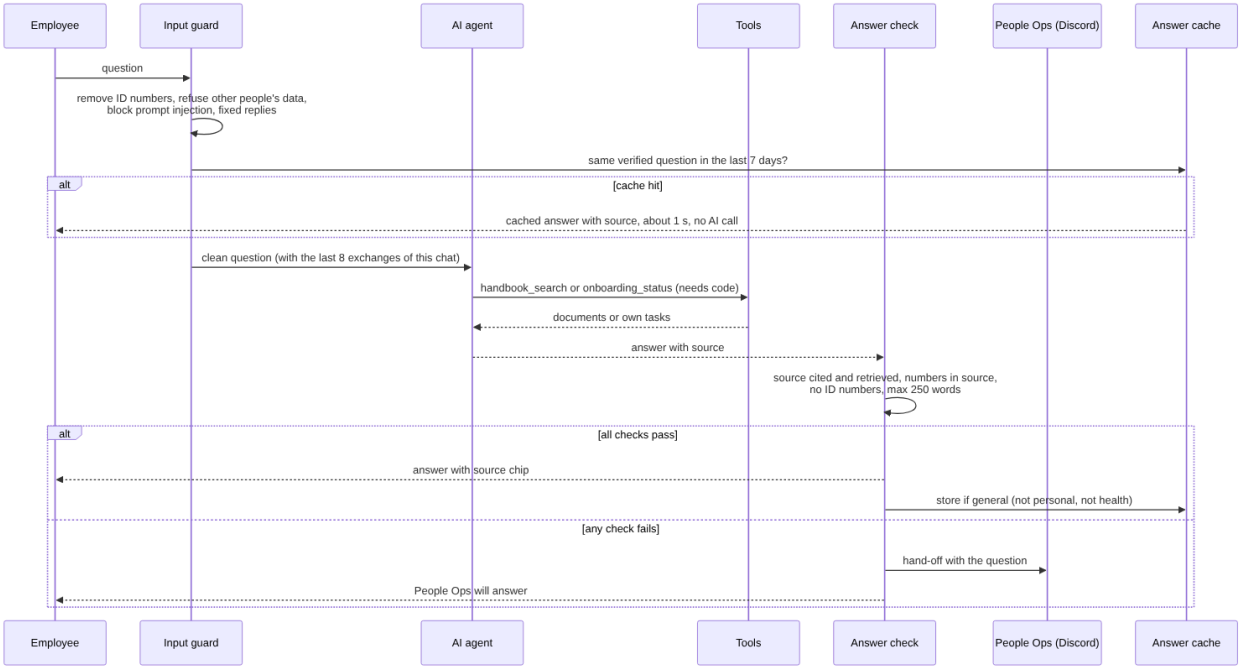
Three flows share one database.

Flow	Path
Onboarding	HR portal (10) → `hris_employees` → Workday report API (9) → Workday sync (8) → Intake (1): rules, AI draft, code check → HR approval in Gmail → welcome e-mail, laptop, account, Discord tasks, Jira tickets for IT and Facilities through the Jira MCP server (create issue only)
Payroll data	Daily scheduler (4) → single-use form link e-mail → Payroll form (11): checksums → `payroll_details` encrypted with the Vault key → Payroll portal (12): decrypt one record, log the view → DATEV hand-over
Company assistant	Notion (17 pages) → Ingestion (2), nightly → `handbook_documents` (about 97 chunks) → Website chat (5) → Assistant (3): input guard, AI agent with two tools, answer check → reply or hand-off to People Ops

HR approval (sequence)

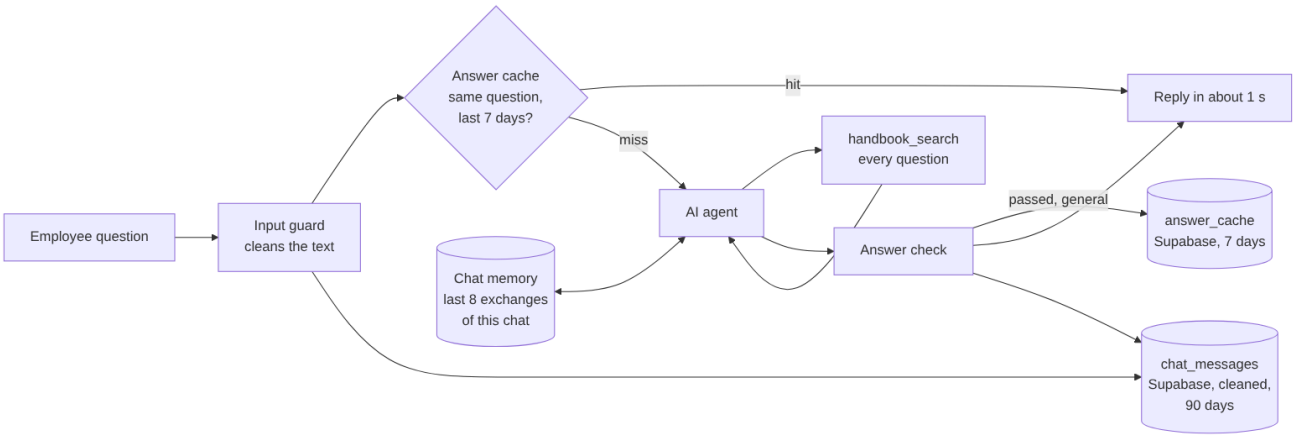


Company assistant (sequence)



Memory, cache and chat storage

The assistant has three kinds of memory. Each has its own purpose and its own limit.



	Chat memory	Answer cache	Chat log
Purpose	Follow a conversation	Answer repeated questions fast, without the AI	Audit, hand-offs, figures for HR
Where	n8n working memory (`Chat memory` node)	Supabase `answer_cache`	Supabase `chat_messages`
Scope	One chat window	Everyone	Everyone
Limit	Last 8 exchanges (8 questions and 8 answers)	Reused for 7 days (`cache_days`), deleted after 30 days	Deleted after 90 days (`chat_retention_days`)
Lost when	n8n restarts, or the user clicks **New**	The knowledge base is rebuilt after a Notion change	Retention clean-up in workflow 4

When the cache answers: the same words as an earlier question (capital letters and punctuation ignored), the earlier answer passed every check and cited a document, at least 4 words and at most 300 characters.

Never cached: hand-offs, health or other sensitive questions, questions with a personal code, answers about someone's own tasks, follow-ups that start with "and", "also", "what about", "und" or "aber", and test sessions (`qc-`).

What the chat log stores: the cleaned question, the reply, the cited source or `handoff: <reason>`, and the flags `handoff` and `blocked`. ID numbers become `[removed: personal data]`, e-mail addresses, phone numbers and codes become `[email]`, `[phone]` and `[code]`, health questions become `[personal health-related question, text not stored]`, and answers about someone's own tasks are not stored. Test sessions are not logged.

Finding from testing: in a long chat the agent sometimes answered from its chat memory without a new search. The answer check could not verify the answer (`no retrieval was recorded`) and handed it off, which is safe but unnecessary. Rule 1 of the system prompt now requires a new search for every question.

Useful queries:

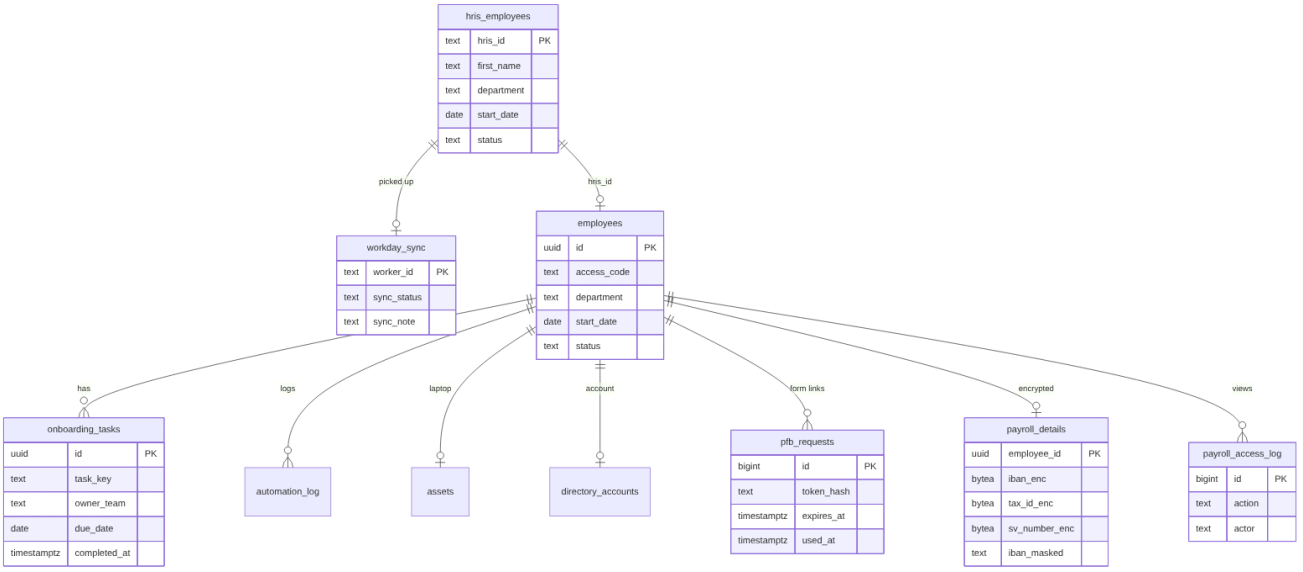
```
-- All chat messages, newest first
select created_at, session_id, user_message, reply, sources, handoff, blocked
from chat_messages order by created_at desc;

-- Hand-offs with the reason
select created_at, user_message, sources from chat_messages
where handoff = true order by created_at desc;

-- Cached answers and how often they were reused
```

```
select question, sources, hits, created_at, last_hit_at
from answer_cache order by last_hit_at desc nulls last;
```

5. Data model



Other tables: `handbook_documents` (chunks with 3072-dimension Gemini embeddings), `chat_messages`, `answer_cache`, `code_attempts`, `quality_checks`, `automation_log`, `assets` (14 seeded laptops), `directory_accounts`.

Views: `onboarding_overview`, `workday_sync_overview`, and seven `reporting.*` views for Grafana (`hires_by_month`, `hire_status`, `task_progress`, `day1_readiness`, `payroll_forms`, `chat_daily`, `automation_daily`).

6. Key design decisions

Decision	Why
Mock Workday report API (workflow 9) instead of the n8n form	In a real company hires start in the HR system. The mock copies the Workday custom report URL and needs a bearer token. Moving to a real tenant changes the URL and credential, not the logic
Only `Successfully Completed` hires start onboarding	Matches the Workday hire business process. `In Progress` waits for HR
Rules in Code nodes, not in the prompt	Readable, testable, same result every time
Gmail send-and-wait for approval	HR approves in the tool they already use. The run pauses until HR clicks
Personal code for the assistant	An employee only sees their own tasks. Lockout after wrong tries
pgcrypto AES-256 with the key in Supabase Vault	Bank and tax data are never stored in clear text and never pass through the AI, Discord, LangSmith or n8n execution logs
HMAC-signed (Hash-based Message Authentication Code) page tokens on HR and payroll portals	n8n serves webhook pages in a sandbox, so origin checks fail. Signed tokens stop forged form posts
Knowledge base rebuild with run id	New chunks are written first, old ones deleted only after success, so the assistant is never empty
OpenRouter for chat, Gemini for embeddings	Gemini free tier rate limits made the assistant slow (24 to 33 s). OpenRouter brought it to about 2.7 s

7. Scaling and production path

Today (capstone)	Production
n8n on a laptop in Docker, public link through trycloudflare	Always-on server in Germany with HTTPS, see Deployment
Basic Auth for HR and payroll	Company single sign-on
Mock Workday API	Real Workday tenant through an integration system user (OAuth)
Free AI tiers	Paid tier with a fallback model
Manual retention	Automatic deletion of payroll details after the legal retention period
The AI decides when to search the knowledge base	A fixed search step before the AI on every question, see Improvement plan
Notion knowledge base	SharePoint Online through Microsoft Graph